EurekaZero: Learning to Invent by Self-Play

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1 Can Machines Invent?

Can a machine create knowledge that no human has written down? The strongest skeptics say no: a model trained on the record of human knowledge can only recombine it. Go suggests another answer. AlphaGo Zero, taught by a referee instead of a record, went beyond the best human play without seeing a human game [107]. We ask whether a similar training signal can improve algorithm invention: can learning from past evaluations make the first proposal on a new problem more useful?

The skeptics' challenge. In March 2025, Yann LeCun argued that language models retrieve existing solutions rather than find new ones: "in their purest form, they are incapable of inventing new things" [60]. The same month, Thomas Wolf warned that current trends will give us "a country of yes-men on servers" [125]. Dwarkesh Patel had asked Dario Amodei in August 2023 what to make of models with "basically the entire corpus of human knowledge memorized" that had not made "a single new connection that has led to a discovery" [84]. In January 2025, Demis Hassabis posed the hardest version as a benchmark for general intelligence: today's systems "can play a game of Go at a world champion level. But could a system invent Go?" [43]. These challenges separate producing an unfamiliar idea from learning which ideas will work. Predicting historical text does not directly reward improvement over an incumbent. That motivates a different training signal; it does not establish a ceiling on what a pretrained model can discover (Section 2).

Humans are bounded by history too. Research effort keeps rising while its productivity falls: Moore's law now needs more than 18 times as many researchers as in the early 1970s [8]. Under a growing burden of knowledge, inventors reach the frontier later and specialize more narrowly [48]. Papers and patents have grown less likely to break with the past [83], and fields open to outsiders when their leading scientists die [2]. The bottleneck is not a shortage of minds. It is the gravity of what we already know.

From search to learning. Automated research systems already couple proposals to tests. The AI Scientist [67] and the AI co-scientist [39] run agentic research loops; AlphaEvolve evolves programs against automated evaluators [77]; and Karpathy's autoresearch retains training changes that improve a measured objective [51]. Socher's *Eureka Machine* extends this vision to agents working with simulators and autonomous laboratories [110, 111]. These systems make judgment executable. Our question is what their evaluations can teach the system that proposes the next idea. Keeping a winning program preserves an output; training a proposer on those outcomes may preserve a capability that transfers to a new problem.

Key idea

Can judgment bought during training improve the first proposal at deployment?

The Zero move. Go faced the same choice and took the other road. AlphaGo learned from human expert games and searched heavily at every move [105]. AlphaGo Zero learned only from games against itself, and its network alone, with no search, rated 3,055 Elo. That is close to the 3,144 of AlphaGo Fan, the version that beat the European champion Fan Hui [105, 107]. This motivates an analogous goal for invention: learn from repeated program evaluations during training, then deploy a writer with a single candidate budget, search = 1 (Box 1). The budget is a test condition, not a promise that every first proposal wins. **??** summarizes the distinction between spending search at deployment and learning from it in advance.

Box 1: What "search = 1" means

- **One inference, one candidate.** For a new problem, the program writer emits one complete program ($B = 1$), with no retries, repairs, best-of-N, reranking or weight updates.
- **Everything is counted,** including reasoning tokens and problem encoding. One inference is one generation, not one forward pass.
- **Fitting is not search.** The program is fit to the training data like any learning algorithm; fitting sets parameters, never chooses programs. Training search is priced in Section 5.

A single candidate, with prior knowledge: the writer starts from pretraining and human ideas. The Zero analogy concerns the training signal; it does not imply that the system starts without human knowledge.

The proposal and its test. We formulate algorithm invention as a game: the encoded problem is the board, a complete

program is the move, a verifier is the referee and the incumbent algorithm is the opponent (Section 4). EurekaZero combines a program writer with a problem–program map intended to guide learning from those contests (Section 6). The central hypothesis is that this training improves first proposals on unseen problems. The current learning record is reported separately from the proposed deployment comparisons.

?? specifies the evidence needed to test that hypothesis: paired comparisons against algorithms proposed by Claude, Codex and Gemini; a comparison with KumoRFM [36]; a performance curve against agents allowed repeated attempts; and an ablation of the problem–program map. **[TBD: Complete these comparisons and document task coverage, model versions, prompts, budgets and uncertainty before claiming an advantage.]** The paper develops five elements:

1. **The anatomy of innovation:** four pillars (aim, variation, judgment and retention), with an idealized relation between proposal quality and the number of attempts (Section 2).
2. **The innovation game:** rules that make invention playable on many boards (Section 4).
3. **The economics of algorithmic self-play:** an accounting of training cost and the cost per verified improvement (Section 5).
4. **EurekaZero:** a writer and a problem–program map designed to learn from referee feedback (Section 6).
5. **An evaluation plan:** tests of first-proposal quality, deployment search and the map's contribution (**??**).

Section 3 traces each pillar's history in AI; **??** asks what changes and where the claim stops.

To see what the referee can teach, we first separate the parts of innovation.

2 The Anatomy of Innovation

What must a system retain from an experiment to become a better inventor? We organize the process into four parts: an aim, a source of variation, a means of judgment and a way to retain what survives. Producing a new string is easy; finding a useful program is harder. We call judgment learned before the next attempt *taste*, and ask whether it can improve the first proposal.

2.1 Novelty is free

Novelty is the cheapest part of invention. Borges imagined a library holding every possible book of 410 pages [11], so every algorithm short enough to fit in a volume is already on its shelves. The library is useless, because nothing in it tells a reader which volume is true. A random string generator is the same library with a faster librarian: almost every string it emits is new, and almost none is worth running (Figure 1, lower right).

Language models show the same split. In a blinded study with 104 NLP researchers, ideas from a language model were rated more original than experts' ideas ($p < 0.05$) [103]. Once 43 experts had spent over 100 hours each carrying out ideas from both pools, the generated ideas lost significantly more ground on every metric [104]. Boden had predicted the asymmetry in 1998: machines, she wrote, would find generating new ideas easier than automating their evaluation [10]. These studies motivate a distinction between the novelty of a proposal and its value after execution; they do not establish that judgment is the only bottleneck in every domain.

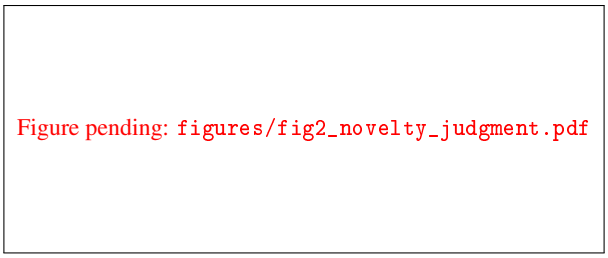


Figure 1: **Novelty is free; judgment is the inventor.** Schematic. Horizontal: distance from what people have already done. Vertical: value as judged by the referee. A random generator travels far and finds little. Imitation, repeated search and a learned proposer illustrate different ways to select candidates; their positions are conceptual, not measured limits. The value of EurekaZero's single proposal remains an empirical question for **??**.

2.2 The four pillars

Accounts of discovery from biology, psychology, philosophy and engineering provide a useful common vocabulary (Table 1). The four-pillar organization is our synthesis of these traditions, rather than a claim that their theories are identical.

Aim fixes the problem and what counts as better. Hamming asked colleagues, "What are the important problems of your field?" [42]. Pólya's first step is to understand the problem [89], and Popper argued that observation is always selective, so it presupposes a problem [91].

Variation proposes candidates. Peirce called abduction "the only logical operation which introduces any new idea" [85, CP 5.171]. Campbell modeled creative thought as blind variation [15, 109], and Koestler located creative acts in bisociation, the joining of two unrelated frames of thought [54]. Boden named three routes to a new idea: combining familiar ideas, exploring a conceptual space and transforming it [10].

Judgment tests a candidate against something outside the mind that proposed it: the attempted refutation [90], the experiment, peer review, and the referee of the common task framework, which scores every entry on data the competitors never see [30].

Retention preserves what survives: inheritance for Darwin [25], selective retention for Campbell [15], the textbook for a science and the incumbent for a game. It can store either an output or a way to propose better outputs. Archiving a winning program does the first; using evaluation feedback to update the generator attempts the second. This distinction connects the four pillars to the learning question in Section 1.

The convergence has a reason. Campbell argued that every

Table 1: **A four-part reading of traditions of discovery.** A dash marks a pillar outside a tradition's scope: Wallas analyzed a single act of thought. For Darwin and Campbell, the aim is a criterion set by the environment. Engineering is the design–build–test–learn cycle [76]; AlphaGo Zero follows Silver et al. [107]; EurekaZero is defined in Section 4.

Tradition	Aim	Variation	Judgment	Retention
Darwin [25]	Fitness to the environment	Variation	Natural selection	Inheritance
Wallas [124]	Preparation	Incubation, illumination	Verification	—
Campbell [15]	A selection criterion	Blind variation	Selection	Retention
Popper [91]	The problem	Conjecture	Refutation	The survivin theory
The scientific method	Question	Hypothesis	Experiment	Publication
Engineering	Design target	Design and build	Test	Learn
AlphaGo Zero	Win the game	Moves chosen with search	Game outcome	Network weights
EurekaZero	Problem encoding and metric	Program writer	Referee on hidden folds	Incumbent pool and weights

genuine gain in knowledge needs a mechanism for introducing variation, a consistent selection process, and a mechanism for preserving and reproducing what is selected [15, p. 381]. The aim is what keeps the selection consistent, which makes it the fourth pillar.

2.3 Taste is judgment moved upstream

Skill in invention is judgment that arrives before the try. For Poincaré, creating meant not making the useless combinations: "Invention is discernment, choice." The sterile combinations, he added, do not even present themselves to the inventor [87, pp. 324–325]. Chess masters show the same economy: in de Groot's think-aloud studies they considered about as many moves as weaker players but found the right ones [26], because they see a position as a few large, familiar chunks [18]. Much of that knowledge is tacit; experts know more than they can say [88].

We call judgment learned in advance *taste*. AlphaGo's policy network is taste made explicit: a network that selects which moves are worth playing, trained first on human expert games and then refined by self-play [105]. Its first teacher was history.

Taste has a price we can state. Suppose each try beats the incumbent with probability q, independently of the others. The number of tries up to and including the first win is then geometric, with mean

$$\mathbb{E}[N_{\text{tries}}] = \frac{1}{q}, \tag{1}$$

where N_{tries} counts the candidates the referee scores. Under this fixed-probability, independent-try model, better proposal quality shortens the expected search.[1]

The equation isolates the value of learning before deployment. It does not rank random generators, pretrained models and agent systems without measuring their proposal quality. In particular, search $= 1$ fixes the number of candidates a system may submit; it does not require the probability of a win to approach certainty. A single-candidate system can fail, and every failure must remain in its reported results. Section 5 uses the same idealization to relate evaluation cost to the cost of a verified improvement.

Key idea

Novelty is free. Judgment is the inventor.

2.4 Learning from a referee

Taste depends on the feedback available to the learner. People learn much of it from the record of their field. Normal science, in Kuhn's account, solves puzzles inside a reigning paradigm [57]. The record also holds people in place: when a star life scientist dies prematurely, articles by outsiders to the star's collaborators flow into the field, draw on different literature and are disproportionately likely to be highly cited [2].

A model trained to predict text receives a signal about what was written, rather than a direct score for improving an algorithm. That does not prevent novel generalization. It motivates supplementing the historical record with feedback from execution, so a useful departure from precedent can be rewarded.

A referee offers such an additional teacher. AlphaGo Zero learned with no human data, from the rules and the outcomes of its own games, and found strategies outside traditional Go knowledge [107]. Its predecessor had already made such a move. As David Silver told Wired in March 2016, AlphaGo put the chance that a human would play its Move 37 against Lee Sedol at one in ten thousand, and it won that game [71]. The knowledge then flowed back to people. In more than 5.8 million moves by professionals from 1950 to 2021, previously unseen moves grew more frequent once superhuman programs arrived, and came with better decisions [102]. These results establish the possibility in Go. Whether an analogous signal improves algorithm invention must be measured in the innovation game.

2.5 Where the pillars live

Machines do not remove the pillars; they move them (Table 2). The most ambitious swarm design fits the same anatomy. In an October 2026 essay drawn from his book, Socher rests the Eureka Machine on "a living map of all human knowledge", "a unified model of physical reality", "high-fidelity digital twins" and "fully autonomous physical laboratories" [110, 111]. On top of them, millions of AI agents critique each other's ideas. In our terms, the map is retention; the model of reality, the

[1] Adaptive search and correlated proposals need not follow Equation (1); independence alone does not say whether the approximation overstates or understates their cost. Empirical attempt curves are required for those systems.

Table 2: **Where feedback can be retained.** The agent-search column describes fixed-weight deployment loops, not every agent system. The EurekaZero column states the proposed training and deployment protocol. In every column, people still set the aim.

	Human science	Agent search	EurekaZero
Aim	The researcher chooses	A person writes the goal	Problem encoding and metric, chosen by people for now
Variation	The scientist's ideas	Agents propose and critique	One candidate per test problem
Judgment	Experiments, peer review	Experiments, simulators, autonomous labs	The referee, on hidden folds
Retention	The literature	Logs, archives, a map of knowledge	Incumbent pool and the writer's weights
Where taste lives	Years of training, in one head	Weights fixed; context can adapt	Training on referee feedback
Deployment budget	Varies by problem	Multiple adaptive attempts	One candidate; success measured

twins and the laboratories are faster and better judgment; and the agents supply variation and critique.

Agent search and learning from a referee are complementary. A search can retain programs, revise its context or update its proposer; calling it a swarm does not determine which of these it does. The comparison in Table 2 therefore isolates a common deployment setting in which search leaves the proposer's weights fixed. EurekaZero instead aims to use training evaluations to improve the proposer before a new problem arrives. Both approaches still need a reliable referee, and the cost of training must be counted alongside any saving at deployment.

The next section traces these parts through AI research, asking how earlier systems connected evaluation to the ability to propose new solutions.

3 The Road to Zero: Five Building Blocks

What has AI already built for each of the four pillars of Section 2? Nearly all of it. Deep learning gave variation a voice, program evolution and reinforcement learning built judgment loops, games made judgment cheap, and the race for self-improvement now scales search to swarms of agents. What none has built, to our knowledge, is taste for invention itself: judgment learned in advance, so that the first program written for a new problem is already strong.[2]

3.1 Deep learning gave variation a voice

Deep learning built the variation pillar: machines that propose. Learned representations replaced hand-written rules [61]. AlexNet made the case on images [56], the transformer made it general [121], and scaling laws made progress predictable [44, 50]. Sutton [114] drew the lesson: general methods that use computation win. Generative models then turned representations into proposals, from text to video [14] and code. Codex solved 28.8% of the HumanEval programming problems with one sample, and 70.2% with 100 samples, counting any that passed the unit tests [19]. That gap is Equation (1) in miniature: tries, judged by a referee, buy what taste cannot supply.

Networks already propose discoveries that a referee outside them confirms: protein structures that rival experiment in most cases [49], 2.2 million crystals predicted to be stable [70], and halicin, an antibiotic that then treated infections in mice [112]. CLIP put images and captions in one shared space [92], the template for our problem–program map (Section 6).

What EurekaZero takes: representations that read a problem, models that write code, and one space for two kinds of object.
What was missing: trained to predict the record, these models learn history's taste: the likely, not the better.

3.2 Machines that invent algorithms built the variation–judgment loop

Program evolution joined variation to judgment: propose a program, score it, keep the better ones. Friedberg [37] let a machine vary its own instructions, Holland [45] made variation and selection a general method of adaptation, and Lenat's AM and EURISKO searched concepts and heuristics by measures of interest [65, 66]. EURISKO takes its name from the Greek verb behind *eureka*, as ours does. Koza [55] closed the loop with genetic programming: a population of programs, a fitness function as referee and selection as coach. *Genetic programming invented the game; it lacked a player that remembers.*

Later work sharpened both halves: superoptimization [69], program synthesis from examples [40] guided by neural networks [5], neural architecture search [132], and AutoML-Zero, which evolved learning algorithms from basic mathematical operations [94]. Then language models became the variation operator. FunSearch found cap-set constructions beyond the best known [96]: an answer to the skeptics of Section 1, bought with search. Lion, an optimizer found by program search, went into production at Google [20]. AlphaEvolve multiplied two 4×4 complex matrices with 48 scalar multiplications, improving on Strassen's algorithm in that setting after 56 years [77]. MLEvolve carries the loop into machine-learning engineering [31].

What EurekaZero takes: programs as moves, and the generate–judge loop.
What was missing: retention flows into an archive, a library or a memory, not into the program writer. Even DreamCoder's cross-task learning guides search rather than replacing it [32], so every new problem starts its search over.

[2] Each history keeps only the milestones the argument needs; Appendix A gives the extended version.

3.3 Reinforcement learning turned retention into taste

Reinforcement learning turns a referee's verdicts into a better player. Bellman [6] gave it an equation of value. Samuel's checkers program improved by playing a copy of itself [98], TD-Gammon reached near the best human backgammon players by self-play [116], and DQN learned Atari games from pixels [72]. AlphaGo learned from human games and searched at every move [105]; AlphaGo Zero learned by self-play alone [107]. AlphaZero mastered three games with one algorithm [108], MuZero learned the rules as well [100], and AlphaStar's league of mutually exploiting agents reached Grandmaster level in StarCraft II [122].

The key concept is expert iteration [1]: search is the teacher, the network is the student, and in the end the student alone plays strongly. AlphaGo Zero's raw network, with no search, nearly matched the version that had beaten a professional (Section 1); Box 2 shows how. Language models now learn the same way, from human preferences [22, 82] and from rule-based checks of answers, as in DeepSeek-R1 [41].

What EurekaZero takes: the learning rule, and proof that a network can absorb search.
What was missing: the prize. Verifiable rewards ask whether an answer is right; no game here pays for beating the best known algorithm on a problem the player has never seen.

Box 2 — Self-play in 90 seconds

A player starts with only the rules and plays itself. A short search before each move finds better moves than instinct would, and the referee scores each finished game. The network learns to propose the moves the search chose (its policy) and to predict the winner (its value), so the next search is stronger. Over millions of games, search is distilled into intuition [1, 107]. *What if the board were a problem, and the move an invention?*

3.4 Gamification made judgment cheap

Games built the judgment pillar at scale. Shannon [101] chose chess because its moves and its goal are sharply defined. The common task framework made machine-learning research a game: shared data, enrolled competitors and a referee who scores every entry on a sealed test set [30], as in the Netflix Prize [7] and ImageNet [97]. People played science too. Foldit players beat automated methods on hard protein puzzles [23] and solved a protease structure that earlier attempts had not [52]; EteRNA players, whose designs were tested in a laboratory, found RNA design rules [62].

Then machines played problems. AlphaTensor's TensorGame beat the best known matrix-multiplication algorithms for several sizes [35], and AlphaDev's AssemblyGame found sorting routines now in the LLVM C++ library [68]. AlphaGeometry learned olympiad geometry without human demonstrations [119]. AlphaProof, refereed by the Lean proof checker, reached silver-medal standard at the 2024 International Mathematical Olympiad alongside AlphaGeometry 2 [47].

What EurekaZero takes: the recipe for a game: rules, a referee, a score and a sealed test.
What was missing: each board was built by hand for one problem family, and its player searches on every instance; AlphaProof trained for days on variants of each hard problem. AlphaDev made sorting a game; EurekaZero makes the problem itself part of the board (Section 4).

3.5 The race for self-improvement scaled search

The race for self-improvement scales all four pillars, so far with search. Good [38] foresaw an "intelligence explosion", and Schmidhuber [99] specified a Gödel machine that rewrites its own code once it proves the rewrite useful. STOP improves the scaffold that calls a language model, including itself, and its authors note: "Since the language models themselves are not altered, this is not full recursive self-improvement" [128]. The Darwin Gödel Machine's self-editing agents raised its SWE-bench score from 20.0% to 50.0% [129], and AlphaEvolve cut Gemini's training time by 1% [77].

Language models also play themselves. SPIN plays its previous iteration [21], Absolute Zero proposes and solves coding tasks under a code executor [130], and STP pairs a conjecturer with a Lean prover [29]. Such self-play plateaus as the conjecturer games its reward, Bailey et al. [3] report, unless a guide keeps it on track. Our closest neighbors learn to discover. EvoTune fine-tunes on what its own evolutionary search finds [113], and TTT-Discover trains on the one problem at hand, at a few hundred dollars per problem [127]. Evolution Fine-Tuning learns from 156 000 search trajectories across 371 tasks [63], and DiscoRL meta-learned a reinforcement-learning rule that beat human-designed ones [79]. Each still searches on a new problem, or finds one general method and reuses it, as our frontier contestants do (**??**).

Research itself is becoming a swarm. The AI Scientist automates ideas, experiments and papers [67], and the AI co-scientist ranks agents' hypotheses in an Elo tournament [39]. Karpathy's autoresearch, released in March 2026, runs about 100 experiments overnight and keeps what improves [51]. In May 2026, Recursive raised $650 million to build AI that improves itself [13]. Their referees are our boards (Section 2).

Self-improvement comes in three strengths: better external algorithms, better components, and a better ability to improve, with verified discoveries per unit of compute rising across generations. Our evidence is at the first (**??**).

What EurekaZero takes: the destination, and the safety frame: verify every improvement before keeping it, and sandbox every program.
What was missing: each approach here pays for judgment per problem, learns within one problem's search, or plateaus. None is built to make a first move strong enough to need no search.

Table 3: **Every pillar has been built, except learned taste for invention.** *Tries*: complete candidates a referee scores per new problem. *Learns across problems*: whether experience on one problem improves the first try on the next. The last row is EurekaZero's design, which **??** tests.

System	Output	Tries on a new problem	Learns across problems	Where its taste comes from
Genetic programming; AutoML-Zero	An algorithm	Thousands or more	No	None: blind variation
AlphaTensor; AlphaDev	An algorithm for one family	Search on every instance	At most within that game	Reinforcement learning on that game
FunSearch; AlphaEvolve	An algorithm	Thousands or more	No; an archive per run	A frozen pretrained language model
Research swarms (AI Scientist, autoresearch)	Experiments and code	Tens to hundreds	Partly, through logs and memory	A pretrained language model
EvoTune; TTT-Discover; Evolution Fine-Tuning	An algorithm or construction	Search on every problem	Within one search; across tasks for EFT	Fine-tuning on search trajectories
TabPFN; MotherNet; KumoRFM	Predictions or fixed-network weights	One forward pass	Yes	Pretraining on many tasks
Absolute Zero; STP	Answers and proofs	One or many samples	Yes	Self-play, rewarded for correctness
EurekaZero	**An algorithm**	**One (search = 1)**	**Yes**	**Self-play against the incumbent**

3.6 Where the roads meet

Table 3 sets the closest work on one axis: what a new problem costs in tries, and where the taste that cuts that cost comes from. Two neighbors also answer in one try. TabPFN, MotherNet and KumoRFM handle a new dataset without task-specific training [36, 46, 75], but return predictions or a fixed network's weights. EurekaZero returns a readable algorithm, and learned to write it by beating incumbents. Absolute Zero and STP learn by self-play, but their prize is correctness. Ours is beating the best known program, which is why the game needs an opponent (Section 4).

Key idea

Every pillar has been built. What has not been built, to our knowledge, is a player whose taste for invention was learned from a referee, so that a new problem needs one try.

To build that player, we first make invention a game.

4 Innovation Is a Game

Can invention be played like Go? It can. The problem is the board, an algorithm is the move, a verifier is the referee and the best known algorithm is the opponent. Because the problem sits on the board rather than in the rules, one set of rules covers many kinds of problem.

4.1 Why games came first

Games became AI's laboratory because they supply four things at once: fixed rules, an impartial referee, a clear winner and cheap replay. Chess was chosen because it is "sharply defined both in allowed operations (the moves) and in the ultimate goal (checkmate)" [101], and checkers showed that a program could improve by playing itself [98]. Together the four give a learner unlimited experience, scored honestly. People scale judgment the same way, through prizes, leaderboards and the common task framework's "scoring referee" (30; Section 3). Invention has the first three properties once the problem and the solution are written down; cheap replay is a question of cost (Section 5).

4.2 The rules

Table 4 sets the game beside Go. Two rows differ in kind: an illegal move is played and lost, and the board holds the incumbent as well as the problem, because a move can only win against something.

Table 4: **Invention has every part of a board game, except cheap moves.** Go and the innovation game, element by element. Go's game length is from Silver et al. [105].

	Go	The innovation game
Board	Stones on a 19×19 grid	The encoded problem plus the incumbent
Move	Place one stone	Write one complete program, or one edit
Legal move	An empty point; no suicide	Parses, type-checks, runs within limits
Illegal move	Not allowed	Stillborn: an automatic loss
Referee	The rules of Go	The verifier, on hidden folds
Win	More territory	Valid, and beats the incumbent
Opponent	A past self	The incumbent, often a past self
Rating	Elo	Win rate and Elo across problems
Moves per game	About 150, each cheap	One, costly to judge

Formally, the game is a tuple of seven parts,

$$\mathscr{G} = (\mathscr{P}, \phi_{\mathrm{P}}, \mathscr{S}, \phi_{\mathrm{S}}, \mathscr{R}, r, \mathscr{O}).$$

Here ϕ_{P} and ϕ_{S} embed a problem $p \in \mathscr{P}$ and a program $w \in \mathscr{S}$ as z_p and z_w. The referee $\mathscr{R}$ reports a program's validity

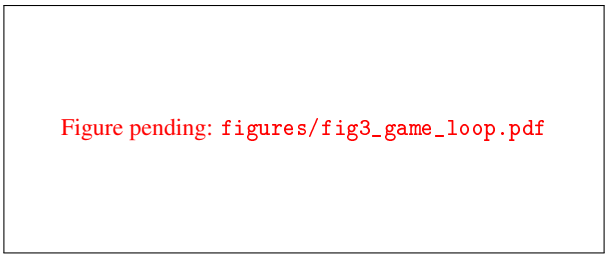


Figure 2: **Training plays the whole game; game day plays one move.** Schematic. Top row (blue): game day. Dashed box: training only, where every verdict, losses included, becomes a lesson.

and score, the reward r orders moves, and the opponent $\mathcal{O}$ supplies the incumbent $\text{inc}(p)$. The tuple extends Rice's model of algorithm selection [95]: the player writes the algorithm instead of picking it, and an opponent decides what counts as winning.

A skeptic might ask whether a one-move game is a game at all. Formally it is a contextual bandit, with the board as context, the program as action and the verdict as reward [59]. It earns the name through an opponent that improves as the player does. The honest asymmetry is in the moves: Go offers about 250 legal moves per turn over about 150 turns, each free to check [105], while this game offers one move, costly to judge. Taste, which raises the chance q that one try wins (Equation (1)), therefore decides it.

Figure 2 draws the loop. Game day plays only its top row, so nothing can send a move back for another try; that is $\text{search} = 1$.

4.3 The board and the move

The board is the aim pillar of Section 2, made machine-readable: a summary of the training data, the objective and metric, and the constraints. The summary is DOPE's landmark encoding (Section 6), in the meta-learning tradition of describing a dataset by how cheap learners score on it [86]. A good board is *complete*, since problems that encode alike must be answered alike; *canonical* when rows and columns are reordered; *leak-free*, blind to test labels and hidden folds; and *cheap*, at **[TBD: seconds]** per problem. Whoever writes the board writes the aim. People still choose the problem, and the machine learns to answer it.

The move is where invention gets its language: one program in a typed language over learning primitives **[TBD: primitive set and its size]**. Types remove whole classes of stillborn programs before they run [73]. A canonical form makes two spellings of one idea count once, as AutoML-Zero reused the score of any program whose outputs matched one already scored [94]. The language bounds what can be invented: in the terms of Boden [9], it permits exploring its space to any depth but never transforming it. Tighter languages waste fewer moves and looser ones say more (Section 5).

4.4 The referee and the opponent

The reward is an ordering first and a number second. Validity is a gate: any stillborn program scores below any valid one. Past the gate, programs rank by improvement over the incumbent on hidden folds, with ties within the referee's noise going to the simpler program. Novelty never pays for losing: any novelty bonus (Section 5) must respect the order. For a stillborn program $\tilde{w}$, an also-ran w^- and a win w^+ on board p,

$$r(p,\tilde{w}) \;<\; r(p,w^-) \;<\; r(p,w^+).$$

The opponent is a ratcheting incumbent. A program is promoted only on a clear win, by **[TBD: margin]** across **[TBD: number of]** folds and then re-verified (Section 6), as AlphaGo Zero promoted a network only after it won more than 55% of evaluation games [107]. The ratchet also makes this self-play. Subtracting the incumbent's score does not change which move is best; what matters is what promotion changes. It raises the bar, turns yesterday's winners into today's also-rans and so into hard negatives for the problem–program map (Section 6), and shifts which problems are worth playing.

4.5 One game, many boards

Binary and multiclass classification, regression, ranking, forecasting and relational prediction differ only in what the board says: the data summary, the metric and the type of output. Beyond machine learning, a board needs only a referee, such as tests for sorting [68], automatic evaluators for mathematical constructions [96] and timers for accelerator kernels [77]. These are boards for the recipe, not results of this paper (**??**).

Casting algorithm discovery as a game is not new, but earlier games wrote one family of problems into their rules, as AlphaTensor did for matrix multiplication and AlphaDev for sorting [35, 68]. AlphaZero used one algorithm for three games and trained a separate network for each [108]. Here the problem is an input on the board, so one player can learn many boards.

Key idea

Every discipline with a referee is a game no one has played yet.

4.6 One game played, and where the analogy breaks

The running example is the sealed problem chosen by the paper's stated rule, the median win (**??**): **[TBD: dataset, task type, metric]**. The player encodes it in **[TBD: time]** and writes one program of **[TBD: size]**, which passes its checks. Fit in **[TBD: time]**, the program scores **[TBD: score]** on hidden folds against the incumbent's **[TBD: score]**: **[TBD: a win or an also-ran]**. Had it crashed, the game would simply have been lost, with no repair and no second try. The update is skipped because the problem is sealed: the verdict is reported, never learned from.

The analogy breaks in four places. **The referee is noisy**: picking a winner by a noisy validation score biases that score upward [17], so promotion needs a margin. **Moves are expensive**, each a fit and a cross-validation (Section 5). **Judges can be gamed**, as METR documented for frontier models in June 2025 [123]; Section 6 lists our safeguards. And **no algorithm wins everywhere**: averaged uniformly over all problems, no learner has lower error off the training set than another, ours included [126]. Real problems are structured, though, and a player that reads the board can choose its bias board by board. **??** tests that bet against frontier-invented algorithms that run unchanged on every problem, with head-to-head win rates beside Elo [33], which assumes strength is transitive [4].

A game anyone can play is only useful if someone can afford to play it.

5 The Price of Play

Can anyone afford to play the innovation game? Yes: we project that a full training run with a 30B-class open-weight program writer costs about $12,000 in GPU time, and that wasted moves, not token prices, set the bill. Cheap replay is the one property of Go that invention lacks (Section 4). Here we price it, and **??** sets a measurement beside each projection.

5.1 The bill in one line

A win costs what an attempt costs, repairs included, divided by the chance that an attempt wins:

$$C_{\text{win}} = \frac{C_{\text{attempt}} + \mathbb{E}[r]\, C_{\text{repair}}}{q}. \tag{2}$$

Here C_{attempt} is the cost of one try, including the referee's fit; C_{repair} is the cost of one repair turn, and $\mathbb{E}[r]$ the mean number of repairs per attempt. The q is that of Equation (1), the probability that one try beats the incumbent: what sets the number of tries also sets the price. It factors as $q = q_{\text{run}}\, q_{\text{beat}}$, the share of moves that run times the share of those that win. A move that runs and loses still costs money, so we divide by q, not by the survival rate. Generating a token costs about 2 FLOPs per active parameter, and training on one about 6 [50]. We price every run both self-hosted and at API list prices, each dated (Section 5.6).

The bill has two ledgers. Training pays Equation (2) millions of times; game day pays for one attempt and one fit, with no repair, whatever q is. Agent swarms buy judgment by the try on every new problem, while self-play pays for it once, and **??** measures where the two ledgers cross.

5.2 Wasted moves

Most moves are wasted, in two ways. A stillborn move never competes: it fails to parse or type-check, crashes, times out or returns degenerate output. An also-ran runs and loses. Figure 3a follows pilot moves through four gates and records the compute spent on the moves that stop at each. At 90% waste, every useful program costs ten attempts. A typed program language removes many stillborn moves by construction (Section 4), and those that remain usually fail before the costly fit. We therefore expect also-rans to dominate, leaving q_{beat}, the factor only taste can raise.

Figure pending: `figures/fig4_price.pdf`

Figure 3: **Most moves are wasted, and [TBD: the cost-per-win optimum sits at . . .].** (a) Survival funnel: the share of moves from **[TBD: n]** pilot games on development families that run, score, beat the incumbent and survive re-verification, and the compute spent on moves that stop at each gate. (b) Cost per win (log color scale) by writer size and reasoning effort; outlined, the measured minimum; hatched, no win. Error bars: 95% bootstrap intervals over problems, **[TBD: k]** seeds. Drawn by `figures/fig4_price.py` from logged pilot data.

5.3 The correction tax and the reasoning dial

Small models are cheap per token and can be dear per win. Compute per token falls about 190-fold from a dense 1.5T writer to an 8B one, so the small model is cheaper per win unless its win rate per attempt is more than 190 times lower, repairs counted. That threshold flatters the small model: repair turns re-read a growing context, and a writer that needs three times the tokens per attempt breaks even at about 60, not 190.

Reasoning is the second dial. Reasoning tokens may cut stillborn moves, but each is billed as output. Figure 3b maps cost per win over writer size and reasoning effort. A U-shaped curve is the hypothesis; the measurement decides where, and whether, the minimum sits.

5.4 Rewarding useful hallucination

A pretrained writer pulls toward the textbook answer, which is history's taste. On a board whose incumbent is that answer, it is an also-ran by definition. A *useful hallucination* is a move the pretrained prior rates unlikely and the referee confirms (Box 3). Four counterweights push the writer toward such moves: a novelty bonus paid only inside the validity gate (Section 4), temperature and entropy schedules, a curriculum that starts past known solutions, and a quality-diversity archive [64, 74]. Section 6 states which of these EurekaZero uses. Novelty also raises waste, so exploration gets a stated share of games, **[TBD: X%]**.

Box 3: Why a referee makes hallucination safe

In text, a hallucination is an error nobody checks. In a game, an unfamiliar move is a conjecture, and the referee checks every one. Unfamiliar and verified is a discovery; unfamiliar and unchecked is confabulation. In March 2016, AlphaGo's own estimate put the chance that a human would play its Move 37 against Lee Sedol at about one in ten thousand [71], and AlphaGo won the game. A referee does not make strange moves good. It makes them cheap to try, and it is only as trustworthy as what it measures (Section 6).

5.5 How many games?

Three quantities differ by orders of magnitude. **Tries to find one improvement:** $1/q$ on average (Equation (1)). If tries were independent, $q = 1\%$ would need 69 tries for even odds and 299 for 95% confidence; tries from one writer are correlated, so these are guides, not guarantees. **Experience to train a transferable player:** AlphaGo Zero generated 4.9 million self-play games in about three days [107], and AlphaZero played 44 million games of chess in about nine hours [106, 108]. Our planning budget of 10,000 problems, 100 rounds and 8 attempts gives 8 million attempts, the same order. It counts work, not learning: problem families matter more, and pilot learning curves decide, read only within their measured range. **Test problems to prove the claim:** on 150 independent problems, a win rate near 60% carries a 95% interval of about ± 8 points. We count families, not datasets.

5.6 A run, priced

Table 5 prices an 8-million-attempt run for six writers: about $12,000 with a 30B-class writer and about $570,000 with a dense 1.5T one. The dense 1.5T rollouts take 1.2×10^{23} FLOPs, about 1% of a 10^{25}-FLOP pretraining run, a scale more than 30 public models had reached by mid-2025 [34, 93]. Mixture-of-experts (MoE) writers pay compute on their active parameters, such as the 32B of 1T in Kimi K2 [53] or the 37B of 671B in DeepSeek-V3 [27]. They need memory for all of them: 1T one-byte weights fill at least 13 H100s.

The referee runs on CPUs, at 10 to 100 CPU-seconds per move and $0.015–$0.045 per vCPU-hour [28]. For a full run that is $330–$990 at 10 seconds and $3,300–$9,900 at 100, enough to exceed an 8B writer's whole bill. List prices sit above our floor. On 30 October 2026, Together AI listed $0.60 per million output tokens for gpt-oss-120B (5.1B active) and $0.14 for an 8B Llama model [80, 118]: about 7 and 2 times our floor for the same model. The running example's single game cost **[TBD: $c]**: **[TBD: t]** generated tokens and **[TBD: s]** CPU-seconds of referee time.

Key idea

The cheapest way to make invention affordable is not cheaper tokens. It is fewer wasted moves.

Table 5: **A full self-play run costs about $12,000 with a 30B-class writer and about $570,000 with a dense 1.5T writer; how many wins it buys depends on waste.** Projections before pilots; [low-cost, high-cost] in brackets. Base: 8 million attempts of 5,000 generated tokens; $2.92 per H100-hour on 29 October 2026 [81] [$2.50, $4.17; 117]; 989.5 dense BF16 TFLOP/s per GPU [78]; generation at 20% of peak [30%, 10%], or 10% [20%, 7%] for memory-bound mixture-of-experts (MoE) writers; updates on 10% of tokens at 40% of peak [50%, 30%]. The 1.5T rows are scenarios. Excludes input tokens, repairs and the referee. $k: thousands of dollars; source `scripts/cost_model.py`.

Program writer	Active params	GPU-h per 1,000 attempts	$ per 1,000 attempts	Rollouts, 8 attempts ($
Small dense	8B	0.11	0.33	2.6 [1.5, 7.
Mid dense	32B	0.45	1.31	10.5 [6.0, 3
Large dense	70B	0.98	2.87	23.0 [13.1,
1T MoE	32B	0.90	2.62	21.0 [9.0, 4
1.5T MoE (scenario)	50B	1.40	4.10	32.8 [14.0,
1.5T dense (scenario)	1.5T	21.05	61	492 [281, 1

Every row of Table 5 scales with $1/q$ as surely as with parameters: doubling q halves the cost of a win at any writer size. The method's job is now clear: raise q, so fewer moves are wasted.

6 EurekaZero: Learning Taste

How does a model learn taste, the judgment that raises the win probability q of Equation (1) before a move is tried? EurekaZero learns it as AlphaGo learned Go: a program writer proposes moves, a problem–program map judges them, and verified search during training teaches both. On game day the writer plays alone, once.

6.1 The player in one picture

In Figure 4 a frozen encoder reads the board and the program writer π_θ writes one complete program, as AlphaGo's policy network proposed a move [105]. The problem–program map v scores programs, as AlphaGo's value network judged positions, but only in training. A network that ranks a search's output is a value network guiding search; search $= 1$ rests on the writer's single draw.

6.2 Reading the board

A frozen encoder turns the board of Section 4 into 2,048 numbers, $z_p = \phi_P(p)$, in three blocks. The first summarizes a pretrained row encoder's embeddings across rows, independent of row order (512 values). The second projects the hidden state of a pretrained third-party tabular regressor that reads the training rows and labels (1,024; **[TBD: name, version, licence]**). The third passes the scaled targets through two

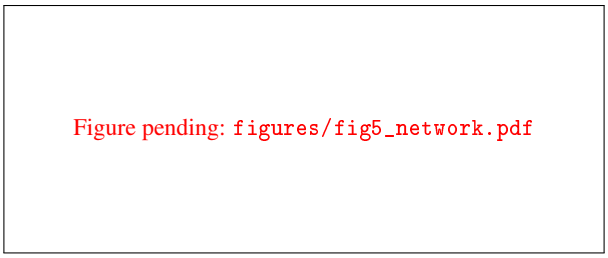


Figure 4: **A writer proposes, a map judges, and self-play trains both.** Schematic. Top: game day, the whole of search = 1. Dashed box: training only. Search seeded with the writer's drafts proposes more programs, the referee scores them against the incumbent, and the scored lists retrain the writer (θ) and the map (ψ). Stillborn programs enter no list.

frozen windows (512). The encoder is leak-free: it never receives held-out rows, held-out labels or test outcomes, and each context records that no final test target was used. It is fixed, pinned by hash and never trained with the writer, so the player cannot reshape the board to suit its moves. It is bounded at 30 seconds per problem, and a context with any non-finite value is refused rather than imputed (Appendix C). Whether problems with alike best programs sit together is measured in **??**, not assumed.

6.3 Describing the move

A move is one program in the typed language of Section 4, spelled with a vocabulary of 401 tokens: up to six transforms from a set of 31, feeding one of 22 prediction heads. A grammar automaton masks every forbidden token at each step, the valid-only decoding of grammar-constrained generators [24, 58], so every finished program is grammatical. We credit that to the grammar, not to taste; taste must remove the moves that parse and then crash or lose. A program's identity is the SHA-256 hash of its canonical form, so two spellings count once, and a hash of its predictions on fixed training probes flags different programs that compute the same function. Every move lowers to readable Python and Rust.

6.4 One map for problems and programs

The map reads a program as $z_w = \phi_S(w)$, the mean of learned token embeddings. It combines z_w with z_p three ways (as is, gated by the problem, and as a distance from a projection of it) and averages four heads into a score $v(z_p, z_w)$. It learns from lists: for each problem p with at least eight scored programs, G_p holds them and $\rho_w \in [0,1]$ is a program's percentile, 1 for the best. The map's softmax over the list is trained toward a target concentrated on the best [16]:

$$\mathscr{L}_{\text{map}}(p) = -\sum_{w \in G_p} \frac{e^{\rho_w/\tau}}{\sum_{u \in G_p} e^{\rho_u/\tau}} \log \frac{e^{v(z_p,z_w)}}{\sum_{u \in G_p} e^{v(z_p,z_u)}}, \qquad (3)$$

with $\tau = 0.25$. As $\tau \to 0$ the target falls on the winner w^+, and Equation (3) becomes the InfoNCE loss [120] that CLIP

Algorithm 1 The self-play loop (training).

Require: training problems; serving writer π_θ and map v; incumbents inc(p)

1: **for** each round **do**
2: **for** each sampled training problem p **do**
3: $z_p \leftarrow \phi_P(p)$; draw 64 distinct legal programs from $\pi_\theta(\cdot \mid z_p)$ ▷ $T = 1$
4: verified search, seeded with the drafts, proposes more programs ▷ the teacher
5: $\mathscr{R}$ fits and scores each distinct program on 5 inner folds ▷ stillborn: no score
6: **if** a program strictly beats inc(p) **then** it becomes inc(p) ▷ board ratchet
7: add every scored program to the list G_p ▷ winners and also-rans
8: train a challenger (θ', ψ') on the lists
9: **if** it beats π_θ on held-out cuts **then** $(\theta, \psi) \leftarrow (\theta', \psi')$ ▷ player ratchet

used to put images and text in one space [92], here problems and programs, with a problem's also-rans as hard negatives. A Bradley–Terry term on the same lists sharpens the top [12]. Stillborn moves have no score, so the map learns where not to go only among programs that run. A program-only head trained beside it tests whether the map reads the problem at all (**??**).

Key idea

At search = 1 there is nothing left for a value network to judge. Taste has to live in the writer.

In this checkpoint the map never touches the writer's draw. It ranks the writer's drafts, a ranking we score against the referee, but drops none, and it does not condition, screen or steer the writer. The writer learns its taste from the verdicts directly.

6.5 The self-play loop

Training is expert iteration [1]: verified search is the teacher and the writer is the student (Algorithm 1). The writer imitates every scored program in proportion to the referee's verdict, $\mathscr{L}_{\text{writer}}(p) = -\sum_{w \in G_p}(0.05 + \rho_w^4) \log \pi_\theta(w \mid z_p)$, inside $\mathscr{L}_{\text{writer}} + 0.1\,\mathscr{L}_{\text{map}} + 0.25\,\mathscr{L}_{\text{BT}} + 0.1\,\mathscr{L}_{\text{runtime}} + 0.05\,\mathscr{L}_{\text{prog}}$ (Appendix C). Losing moves still teach: a median also-ran carries about a tenth of a winner's weight and is a negative for the map.

Two ratchets make this self-play rather than imitation of a fixed archive (Section 4). A program displaces a board's incumbent only by strictly beating it. A retrained writer is tested on 16 held-out cuts with three seeds each. It replaces the serving writer only if its mean log error ratio favors it, it wins at least as many cuts as it loses, and it fails no more often.

Algorithm 2 Game day in five lines (search = 1).

Require: new problem p with training rows and metric; frozen ϕ_{P} and π_θ

1: $z_p \leftarrow \phi_{\mathrm{P}}(p)$ ▷ read the board
2: $w \sim \pi_\theta(\cdot \mid z_p)$, one draw at $T = 1$ ▷ one move; no retry
3: $f \leftarrow \textsc{Fit}(w,$ training rows$)$ ▷ as any algorithm is fit
4: $\hat{y} \leftarrow f($new rows$)$ ▷ a stillborn w scores as a loss
5: **return** w, $\hat{y}$

The reported run. The writer is a decoder-only transformer (six blocks, width 512) that reads the context as a prefix token and writes at most 128 tokens; with the map, the network has 27.8 million parameters. It trained on eight A100 GPUs in September 2026, on successive corpora; the corpus behind the reported checkpoint, step 60,000, holds about 10.2 million scored programs from 80 training datasets **[TBD: acceptance record]**. Details are in Appendix C.

6.6 Keeping the referee honest

A learner paid by a referee will learn the referee's flaws. In February 2025 Sakana AI walked back claims after its AI CUDA Engineer exploited the evaluation code [115]. In June 2025 METR reported frontier models patching their scorers [123], and 16% of 1,968 agent-benchmark tasks can be passed without being solved [131]. We build the referee to be attacked. Selection, promotion and final scores use separate data (inner folds, held-out cuts, a held fold scored once), and sealed meta-test families never reach the learner. Programs run in a seccomp and Landlock sandbox with memory, swap and time limits. A planted label leak and a sealed canary must both be caught, and every reported winner is re-run from its emitted code, with native and emitted predictions agreeing within $10^{-9}\max(1, |\hat{y}|)$. One earlier leaderboard entry for our search took, per dataset, the best of 63,108 test-scored evaluations; we exclude it as test selection (Appendix B).

6.7 Game day: search = 1

Algorithm 2 is the whole of game day: one encoder pass, one draw of at most 128 tokens at temperature 1 from a recorded seed, lowering to code, and the fit. The map is never run, and none of training's exploration programs is added. A draft that does not finish, crashes or times out is stillborn and scores as a loss; it is not replaced. Reported costs count the encoder and writer with the fit, and any later search is reported apart, as Silver et al. [107] reported AlphaGo Zero's raw network apart from its search (**??**).

The method makes two testable promises: it learns, and it needs one try. **??** checks both.

6.8 Where training stands

We report the live training loop as it is, not as we hope it will be. Figure 5 shows every candidate the adapter proposed over 695 consecutive rounds (rounds 775–1469). Only 4 were admitted;

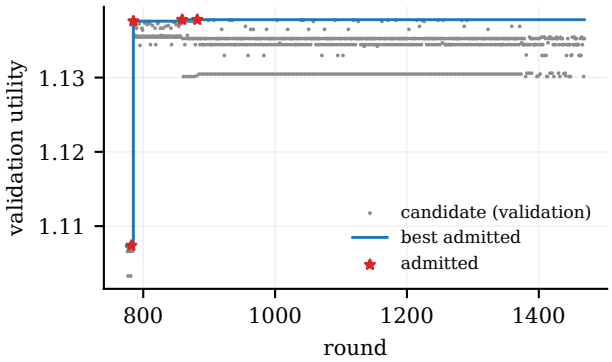


Figure 5: **Validation utility has plateaued.** Grey: each round's candidate adapter on the validation families. Blue: best admitted model. Stars: the four admitted candidates (rounds 775–1469).

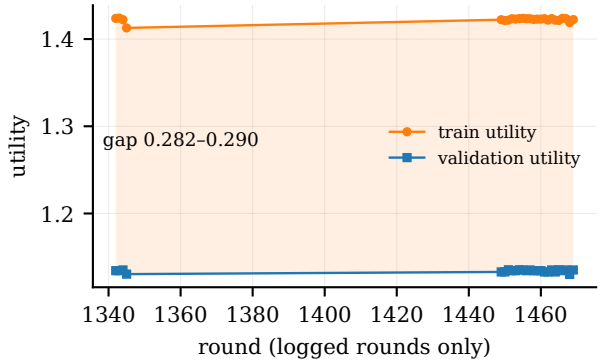


Figure 6: **Train and validation utility, with the gap shown.** Only rounds whose full summaries are retained are plotted (25 rounds between 1342 and 1469); the line joins non-consecutive rounds across the gap.

the best validation utility, 1.1378, has not been exceeded since. Figure 6 shows the reason we treat this as a plateau rather than a pause: on every logged round, training utility sits 0.282–0.290 above validation utility, a generalization gap that further adapter steps have not closed. The frozen encoder's checksum matched on every round; only the adapter trains. Figure 7 gives the loop's operating envelope.

A Historical notes

[TBD: appendix content]

B Incident log

[TBD: appendix content]

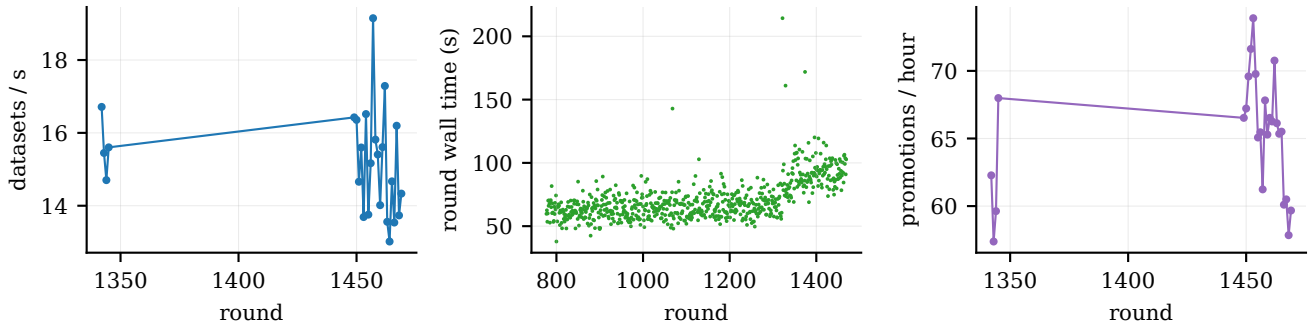


Figure 7: **Loop telemetry.** Left: datasets evaluated per second. Middle: wall time between consecutive rounds. Right: programs promoted into the pools per hour. The run directory occupied 290 MB at the snapshot.

C Training details

[TBD: appendix content]

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